AA—TV wiring v7, PLAN B: NO front driven motors. Deutsch DT connectors only (wet zone); cables epoxy—potted into dryboxes; 14 AWG / 20 AWG.

BUOY (surface, dry)

BT2
USB—C power bank 5 V / 3 A
5V_PI
GND_PI

J1

GPIO12/PWM0		5V	2 — 5V_PI
GPIO13/PWM1		3V3	1
GPIO14/UART_TXD		ID_SD_I2C0/GPIO00	27
GPIO15/UART_RXD		ID_SC_I2C0/GPIO01	28
GPIO16/SPI1_CE2		SDA_I2C1/GPIO02	3
GPIO17/SPI1_CE1		SCL_I2C1/GPIO03	5
GPIO18/SPI1_CE0/PCM_CLK/PWM0			
GPIO19/SPI1_MISO/PCM_FS		GPCLK0/GPIO04	7
GPIO20/SPI1_MOSI/PCM_DIN/PWM1		GPCLK1/GPIO05	29
GPIO21/SPI1_SCLK/PCM_DOUT		GPCLK2/GPIO06	31
GPIO22/SDIO_CLK		CE1_SPI0/GPIO07	26
GPIO23/SDIO_CMD		CE0_SPI0/GPIO08	24
GPIO24/SDIO_DAT0		MISO_SPI0/GPIO09	21
GPIO25/SDIO_DAT1		MOSI_SPI0/GPIO10	19
GPIO26/SDIO_DAT2		SCLK_SPI0/GPIO11	23
GPIO27/SDIO_DAT3			
	GND		6 — GND_PI

Raspberry Pi 4 (buoy)

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power—cycling the Pi USB ports (uhubctl).

TETHER

W1
USB—A — micro—B
5V_PI — GND_PI — + USB data
VBUS_PICO — GND

USB—A to micro—B, ~3 m

Vinyl air tube from the buoy; USB cable runs inside it. USB plugs are dry (buoy, Pico).

PEN1
A1: air—tube fitting (USB inside)

MAIN DRYBOX (upper, on hull)

BOARD A: Pico solder—board

R1 220 Ω: GP12 — M4_PWM
R2 220 Ω: GP13 — M5_PWM
R3 220 Ω: GP14 — M6_PWM
R4 220 Ω: GP15 — M7_PWM
R5 220 Ω: GP16 — THR_PWM
R6 220 Ω: GP19 — PUMP_PWM

220 ohm in series on every PWM line that leaves the drybox.

Raspberry Pi Pico
A1

		VBUS_PICO	39
		VSYS	40
		VBUS	
		+3V3	36
		3V3	
RUN	30		
3V3_EN	37		
GPIO0	1		
GPIO1	2		
GPIO2	4		
GPIO3	5		
GPIO4	6		
GPIO5	7		
GPIO6	9		
GPIO7	10		
GPIO8	11		
GPIO9	12		
GPIO10	14		
GPIO11	15		
GPIO12	16		
GPIO13	17		
GPIO14	19		
GPIO15	20		
		GPIO16	21 — GP16
		GPIO17	22
		GPIO18	24
		GPIO19	25 — GP19
		GPIO20	26
		GPIO21	27
		GPIO22	29
		ADC_VREF	35
		GPIO26_ADC0	31 — LEAK_MAIN
		GPIO27_ADC1	32 — LEAK_REAR
		GPIO28_ADC2	34
		AGND	33 — GND
		GND	38
		GND	3 — GND

TA2
1 — +3V3
2 — GND
3 — LEAK_MAIN
Main leak sensor

All Board A terminals: 5.08 mm pitch (0.2 in, fits the breadboard grid).

TA1

M4_PWM	1
M5_PWM	2
M6_PWM	3
M7_PWM	4
THR_PWM	5
	6
	7
PUMP_PWM	8
LEAK_REAR	9
+3V3	10
GND	11
GND	12

H1 out (20 AWG)

PEN2
A2: H1 cable, epoxy—potted

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / signal board

TB1

M4_PWM	1
M5_PWM	2
M6_PWM	3
M7_PWM	4
THR_PWM	5
	6
	7
PUMP_PWM	8
LEAK_REAR	9
+3V3	10
GND	11
GND	12

H1 in (20 AWG)

TB2

M4_PWM	1
SERVO_7V5	2
GND	3
M5_PWM	4
SERVO_7V5	5
GND	6
M6_PWM	7
SERVO_7V5	8
GND	9
M7_PWM	10
SERVO_7V5	11
GND	12

Servo cable (20 AWG)

TB3
THR_PWM 1 — GND 2
Thruster ESC signal

TB4
PUMP_PWM 1 — GND 2
Pump ESC signal

TB5
+3V3 1 — GND 2 — LEAK_REAR 3
Rear leak sensor

TB6
1 — SERVO_7V5
2 — GND
7.5 V in (14 AWG)

C1
SERVO_7V5 (+) — GND (—)
1000uF 16V

Board B:
— 7.5 V + GND rails: strip of 14 AWG soldered along strips
— No 12 V on the breadboard
— 5.08 mm terminals only

Off—board, in rear drybox

BB1
VBAT 1—8
Bus bars (screw studs, ring terminals):
+ in: D1 pins 1+2 (two 14 AWG)
+ out: buck, pump ESC, D2 thruster
GND bar mirrors the + bar.
+12 V bus bar (>=30 A)

BB2
GND 1—8
GND bus bar (>=30 A)

U1 BUCK
VBAT 1 — VIN+ — VOUT+ — 3 SERVO_7V5
GND 2 — VIN— — VOUT— — 4 GND
12 V —> 7.5 V buck (>=5 A)

U3
VBAT 1 — BAT+ — M+ — 6 PUMP_M+
GND 2 — BAT— — M— — 7 PUMP_M—
PUMP_PWM 4 — SIG
GND 5 — SIG_GND — ESC
Pump ESC (brushed, reversible)

Pump ESC sits dry so any brushed ESC or H—bridge works. BEC wire cut.

PEN3
R2: 20 AWG cables potted (H1, servo, THR sig)

PEN4
R3: 14 AWG cables potted (trunk, thruster pwr, pump motor)

WET ZONE (flooded hull, open rear endcap, splash areas): Deutsch DT connectors live here

20 A ATC, waterproof holder
F1 — VBAT
BT1
3S LiPo (wet, no box)
SW1 — GND
Kill switch, low side, >=30 A

D1A D1B
VBAT 1—1 VBAT
VBAT 2—2 VBAT
GND 3—3 GND
GND 4—4 GND
D1 battery DT06—4S
DT04—4P
D1 battery: 2 contacts per pole = 26 A.
Trunk to rear box: 4 x 14 AWG.

D4A D4B

M4_PWM	1	1	M4_PWM
M5_PWM	2	2	M5_PWM
M6_PWM	3	3	M6_PWM
M7_PWM	4	4	M7_PWM
THR_PWM	5	5	THR_PWM
	6	6	
	7	7	
PUMP_PWM	8	8	PUMP_PWM
LEAK_REAR	9	9	LEAK_REAR
+3V3	10	10	+3V3
GND	11	11	GND
GND	12	12	GND

D4 harness H1 DT06—12S
DT04—12P
D4: main box <—> rear box, 12 x 20 AWG.
Pins 6—7 spare in Plan B.

D3A D3B

M4_PWM	1	1	M4_PWM
SERVO_7V5	2	2	SERVO_7V5
GND	3	3	GND
M5_PWM	4	4	M5_PWM
SERVO_7V5	5	5	SERVO_7V5
GND	6	6	GND
M6_PWM	7	7	M6_PWM
SERVO_7V5	8	8	SERVO_7V5
GND	9	9	GND
M7_PWM	10	10	M7_PWM
SERVO_7V5	11	11	SERVO_7V5
GND	12	12	GND

D3 servos DT06—12S
DT04—12P
D3: 4 servo leads, 12 x 20 AWG.

M4_PWM, SERVO_7V5, GND — M4 PWM — Rudder servo
M5_PWM, SERVO_7V5, GND — M5 PWM — Rudder servo
M6_PWM, SERVO_7V5, GND — M6 PWM — Elevator servo
M7_PWM, SERVO_7V5, GND — M7 PWM — Elevator servo

D2A D2B

VBAT	1	1	VBAT
GND	2	2	GND
THR_PWM	3	3	THR_PWM
GND	4	4	GND
PUMP_M+	5	5	PUMP_M+
PUMP_M—	6	6	PUMP_M—
	7	7	
	8	8	

D2 aft DT06—8S
DT04—8P
D2: 1—2 thruster ESC 12 V (14 AWG), 3—4 ESC signal (20 AWG), 5—6 pump motor (14 AWG), 7—8 spare (2nd pump).

U2 BLDC ESC
VBAT 1 — BAT+ — U — 5 — 1
GND 2 — BAT— — V — 6 — 2
THR_PWM 3 — SIG — W — 7 — 3
GND 4 — SIG_GND
TD1.2 ESC (waterproof, wet)
Factory thruster cable stays on the ESC.

MT1 BLDC
TD1.2 thruster

PUMP_M+ — M3 — PUMP_M—
M3
Ballast pump, 12 V brushed (TBD, <=5 A)

PLAN B: no front driven motors.
No D5, no wheel ESCs, no TB7.
H1 pins 6—7 and Pico GP17/GP18 spare.
Fuse drops to 20 A.

Battery side harness: battery + —> F1 —> split to D1 pins 1, 2; battery — —> SW1 —> split to D1 pins 3, 4. Splits: solder + adhesive heat—shrink. Charge through a DT04—4P —> XT60 adapter.

Deutsch DT: 13 A per contact, 14—20 AWG, IP68 when every cavity is filled (cavity plugs in unused ones). Source side gets sockets (DT06 plug).

Two wire gauges: HIGH = 14 AWG (12 V, 7.5 V, motor), LOW = 20 AWG (signals, 3.3 V, servo leads).
Only connectors: Deutsch DT in the wet zone + 5.08 mm screw terminals / bus bars inside dryboxes.
Two solder—boards (Board A main drybox, Board B rear drybox). Cables epoxy—potted through drybox

PIE AA—TV

Sheet: /
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Title: AA—TV electrical v7, Plan B: no front driven motors

Size: A2	Date: 2026—11—04	**Rev: v7**
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